Appendix B - Survey Instrument (Initial Survey)

The following is the survey content that was hosted on Qualtrics to facilitate user testing. Some elements of the survey (such as HTML elements, hyperlinks, and embedded videos, cannot be displayed in this document, and have been replaced with suitable alternatives.

Context

This survey collects user feedback on an early prototype of a mobile quiz platform designed for use in primary schools. The platform is intended to support short, frequent revision sessions for students by using an easily accessible mobile app. Your tasks today will be based around the web-based application that teachers will use to create and schedule quizzes for the mobile app.

The survey will take approximately 30 minutes to complete and will require you to visit an external website as part of the evaluation. Please attempt this survey on a computer, as the website is not optimised for mobile or tablet devices. You may exit the survey at any time and return later. Your previous responses will be saved, but they will only be accessible if you continue the survey using the same device and browser.

Some key features of the system are not yet implemented. This survey focuses primarily on evaluating the ease of use, clarity of the platform, and overall usability of the current prototype. There are no right or wrong answers. Feedback on confusing or frustrating aspects of the platform is especially valuable.

Background

At which level(s) do you currently teach?

- ☐ Primary 1
- ☐ Primary 2
- ☐ Primary 3
- ☐ Primary 4
- ☐ Primary 5
- ☐ Primary 6
- ☐ Other: _____

How many years of teaching experience do you have?

- ◯ 0–2
- ◯ 3–5
- ◯ 6–10
- ◯ 10+

What subjects do you teach?

- ☐ Science
- ☐ Math
- ☐ English
- ☐ Other: _____

How often do you create revision quizzes or worksheets for use in class?

- ◯ Daily
- ◯ Weekly
- ◯ Monthly
- ◯ Rarely

Scenario Based Usability

Introduction

The platform you are testing has two main interfaces: The teacher-facing web dashboard which facilitates the creation and assignment of quizzes to students, as well as the mobile application, which students will use to access and attempt the quizzes.

In the sections that follow, you will complete 4 scenarios, 3 of which require you to access the web dashboard. All tasks should be done on a computer as the site will not function properly on a mobile phone.

To better evaluate the usability of the platform, the scenarios will not include step-by-step instructions. We'd like you to explore the interface and figure out how to complete each task on your own. A tutorial is available for each scenario, but please only open it if you are genuinely stuck.

Scenario 1

Your teacher login credentials (Qualtrics webservice would send a post request to the credentials issuer here and replace the username and password fields below with the allocated credentials)

Username
${e://Field/teacher_user}

Password:
${e://Field/teacher_pass}

Scenario 1: Class Creation

Your task

Visit the teacher web platform here (hyperlinked to webquizapp.delishad.com, where the prototype dashboard was hosted) and sign in with the provided credentials.

Create a class with the students included in the CSV file below.
You may choose any class name and settings that make sense to you.
After completing the task, scroll down and click next.

Download student list:

Download CSV file (CSV file with unique student names for each teacher, retrieved together with the teacher credentials from the credential issuer.)

View this tutorial only if you are unable to figure out how to do the task

Step 1: Navigate to the Classes section of the platform.
Step 2: Select the "+" button to create a new class.
Step 3: Enter the class name, class level, and optionally, upload an image and change the timezone or colour.
Step 4: Add the students to the class by importing the CSV file.
Step 5: Save the class.

View this video to see the full workflow

(Video is embedded in actual survey)

https://www.youtube.com/watch?v=Pirs8X2CaC8

Scenario 1 Review

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The process of creating and managing a class was easy to understand.	☐	☐	☐	☐	☐
The class setup process was guided clearly by the interface.	☐	☐	☐	☐	☐
The terminology used was familiar and appropriate for a school setting.	☐	☐	☐	☐	☐
Importing student information when creating a class was a straightforward process.	☐	☐	☐	☐	☐
The student account creation process - username and passwords shown to teacher for dissemination, felt suitable for real classroom use.	☐	☐	☐	☐	☐

What parts of the class creation process felt unclear or unintuitive (if any)?

Scenario 2

Scenario 2: Quiz Creation

Your task

The platform allows for the creation of 3 quiz types: Basic, Crossword and Rapid.

In this scenario you will be creating two quizzes:

One Basic quiz
One Crossword quiz

The quiz content will be provided to you. Your task is to create the quizzes on the platform based on the given content. Keep in mind that the provided content is not representative of typical quizzes that would be created on the platform. Instead, it is deliberately short and designed to help you become familiar with the quiz creation workflow.

Please proceed to the next section to view the quiz content that you should use.

Basic Quiz

2.1 Create a Basic quiz

Using the content below, create a basic quiz in the system. After finishing the quiz creation, click next to view the content for the crossword quiz. Quizzes can be created under the "Quizzes" section accessible through the left side bar.

View this tutorial only if you are unable to figure out how to do the task

Step 1: Go to the *Quizzes* section.

Step 2: Click *Create Quiz*

Step 3: Select the quiz type Basic

Step 4: Enter the quiz name, subject, and topic.

Step 5: Increment the question count

Step 6: Edit each question by selecting the question type and typing in the question content.

Step 7: Create the Quiz

This video shows the creation of a basic quiz

(Video is embedded in actual survey)

https://youtu.be/s0EMZG_BT0E?si=dgNbcMYmrgJ4_BF9

(Only one of the 3 basic quizzes below is shown to the user. The quiz shown to the teacher is dependent on the subject that the teacher reported teaching at the start of the survey)

Math — Basic Quiz

Quiz Details

Quiz name: Word Problems
Subject: Math
Topic: Arithmetic
Overall timer: 6 minutes

Context Item

"A shop sells pencils and erasers. A pencil costs $2. An eraser costs $1. Sam buys some items and wants to know the total cost and change."

Question 1: Multiple Choice Item

Sam buys 3 pencils. How much does he pay?

- $3
- $5
- $6
- $8

Question 2: Open-ended Item

Sam buys 2 erasers. How much does he pay?

Accepted answer(s): 2, $2

Question 3: Multiple Choice Item

Sam pays with $10 for items costing $6. How much change does he get?

- $2
- $3
- $4
- $5

English — Basic Quiz

Quiz Details

Quiz name: Reading Comprehension (A Day at the Park)
Subject: English
Topic: Comprehension
Overall timer: 5 minutes

Context Item

"Ben and Mia went to the park after school. They saw a big tree and sat under it. Ben shared his sandwich with Mia. They played on the swing and laughed. Before going home, they picked up litter and threw it into the bin."

Question 1: Open-ended Item

Fill in the blank: Ben shared a _____ with Mia.

Accepted answer(s): sandwich

Question 2: Multiple Choice Item

What did Ben and Mia play on?

- Slide
- Seesaw
- Swing
- Ball

Question 3: Multiple Choice Item

Where did Ben and Mia sit at the park?

- On a bench
- Under a tree
- Near the swings
- On the grass

Science — Basic Quiz

Quiz Details

Quiz name: Plants and Their Needs
Subject: Science
Topic: Plants
Overall timer: 6 minutes

Context Item

Sarah planted a bean plant in a pot near her window. Every morning, she watered the plant and placed it where it could receive sunlight. After two weeks, the plant grew new leaves. One weekend, Sarah forgot to water the plant and left it in a dark room. Without sunlight, the plant could not make its own food, and the leaves were drooping.

Question 1: Multiple Choice Item

Why did Sarah place the plant near the window?

- So it could catch insects
- So it could receive sunlight
- To keep it warm at night
- To protect it from rain

Question 2: Open-ended Item

What process could the plant not carry out without sunlight?

Accepted answer(s): photosynthesis

Question 3: Multiple Choice Item

Which sentence from the context best explains why the leaves were drooping?

- Sarah planted a bean plant in a pot.
- The plant grew new leaves after two weeks.
- She forgot to water the plant and left it in a dark room.
- She checked on the plant again later.

Crossword Quiz

2.2 Create a Crossword quiz

Using the content below, create a Crossword quiz in the system. After finishing the quiz creation, click next.

View this tutorial only if you are unable to figure out how to do the task

Step 1: Go to the Quizzes section and click Create Quiz.
Step 2: Select the quiz type Crossword.
Step 3: Enter the quiz name, subject, and topic.
Step 4: Increment the word count
Step 5: Add the crossword entries (answer + clue) one by one.
Step 6: Generate the crossword
Step 7: Create the quiz

This video shows the creation of a crossword quiz

(Video is embedded in actual survey)

https://www.youtube.com/watch?v=lgjxoLn-dIY

(Only one of the 3 crossword quizzes below is shown to the user. The quiz shown to the teacher is dependent on the subject that the teacher reported teaching at the start of the survey)

English — Crossword

Quiz Details

Quiz name: Interesting Vocabulary Crossword
Subject: English
Topic: Vocabulary
Timer: None

Entry 1

Answer: JOURNEY
Clue: A trip from one place to another.

Entry 2

Answer: WHISPER
Clue: To speak very softly.

Entry 3

Answer: ANCIENT
Clue: Very old; from a long time ago.

Entry 4

Answer: CURIOUS
Clue: Wanting to know or learn something.

Entry 5

Answer: BRAVE
Clue: Showing courage in a difficult situation.

Science — Crossword

Quiz Details

Quiz name: Plants Crossword
Subject: Science
Topic: Plants
Timer: None

Entry 1

Answer: PHOTOSYNTHESIS
Clue: Process by which green plants use light energy to make food from carbon dioxide and water

Entry 2

Answer: TRANSPIRATION
Clue: Loss of water vapour from plant leaves

Entry 3

Answer: CHLOROPHYLL
Clue: Green pigment in leaves that absorbs light energy for photosynthesis

Entry 4

Answer: GERMINATE
Clue: Process by which a seed begins to grow and develop into a new plant

Entry 5

Answer: STOMATA
Clue: Small openings found under a plant's leaves, used for gaseous exchange

Rapid Quiz

2.3 Rapid Quizzes

In addition to basic and crossword quizzes, the platform includes a third quiz type known as rapid quizzes.

Rapid quizzes consist exclusively of multiple-choice questions and are designed with per-question timers, requiring students to respond within a fixed time limit for each question.

On the mobile application, students are allowed to freely navigate between questions in basic quizzes. However, for rapid quizzes, once a student submits an answer to a question, they are unable to return to previous questions. This quiz is designed to test quick recall in a fun manner.

In the interest of time, you will not be required to create a rapid quiz for this activity. Nevertheless, you are welcome to explore this quiz type independently if you are interested.

Scenario 2 Review

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The process of creating a quiz was easy to understand on first use.	☐	☐	☐	☐	☐
The interface guided the quiz creation process in an intuitive manner.	☐	☐	☐	☐	☐
The interface guided me naturally through quiz creation.	☐	☐	☐	☐	☐
The differences between Basic and Crossword quizzes were easy to understand during creation.	☐	☐	☐	☐	☐
The system provided the right amount of configuration options for quiz creation.	☐	☐	☐	☐	☐
Each quiz type (basic, crossword, rapid) was perceived to be clear and distinct.	☐	☐	☐	☐	☐
The available quiz types were aligned with common classroom revision practices.	☐	☐	☐	☐	☐
The available quiz types were sufficient for supporting typical revision needs.	☐	☐	☐	☐	☐

What parts of the quiz creation process felt unclear or unintuitive (if any)?

What features would make quiz creation more efficient?

Are there any types of revision activities you commonly use that are not well supported by the current quiz types?

Scenario 3.1

Scenario 3.1: Quiz Scheduling

Your task
You have created a class and two quizzes in the previous scenario. Assign both quizzes to the class.

Scheduling requirements

Each quiz should start today.
Each quiz should end in 3 days.
Each quiz should have 3 attempts allowed.

View this tutorial only if you are unable to figure out how to do the task

There are actually two ways to schedule the quizzes. One is through the quiz page, and the other is through the class page.

Method 1: Schedule from the quiz page
Step 1: Navigate to the list of quizzes.
Step 2: Click on the schedule button for the quiz you would like to schedule.
Step 3: Select the classes and the dates for which the quiz will be assigned.
Step 4: Increase the attempt count to 3.
Step 5: Assign the quiz.

This video shows the scheduling of quizzes using the quiz page.

Method 2: Schedule from the class page
Step 1: Navigate to the page of that class that you want to schedule quizzes for
Step 2: Click on the scheduling tab
Step 3: Drag and drop created quizzes into the schedule
Step 4: Drag and stretch the quizzes so that it starts today and ends in 3 days.
Step 5: Right click each scheduled quiz and increase the attempt count to 3.

This video shows the scheduling of quizzes using the class page.

(Videos are embedded in actual survey)

Quiz scheduling video (quiz page method): https://www.youtube.com/watch?v=nBkQRau1p5Y
Quiz scheduling video (class page method): https://www.youtube.com/watch?v=CKT7OvRH_pg

Scenario 3.2

Scenario 3.2: Quiz Scheduling

Your task

After assigning both quizzes based on the previous task, adjust the schedule assignment of the basic quiz.

The basic quiz should now end 4 days from today (extended by 1 day)
The basic quiz should now only have 2 attempts.

Adjusting of schedule for a class should be done under the class page

View this tutorial only if you are unable to figure out how to do the task

Step 1: Navigate to the page of that class that you are adjusting the schedule for.
Step 2: Click on the scheduling tab.
Step 3: Adjust the scheduled basic quiz to the correct dates in the calendar.
Step 3.1: Alternatively, right click the scheduled basic quiz and adjust the dates in the menu.
Step 4: Right click the scheduled basic quiz and adjust attempt count to 2.

This video shows the adjusting of a class schedule.

(Videos are embedded in actual survey)

Schedule adjustment video: https://www.youtube.com/watch?v=26VZX_O5Aow

Scenario 3 Review

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The location for scheduling a quiz was easy to identify.	☐	☐	☐	☐	☐
The scheduling process was easy to understand without external instructions.	☐	☐	☐	☐	☐
The interface guided me naturally through the scheduling process.	☐	☐	☐	☐	☐
The system provided sufficient feedback when I scheduled or moved a quiz.	☐	☐	☐	☐	☐
Clear feedback was provided when quizzes were scheduled, moved, or updated.	☐	☐	☐	☐	☐
It was easy to adjust or correct a scheduling mistake.	☐	☐	☐	☐	☐
Rescheduling a quiz was straightforward.	☐	☐	☐	☐	☐
I felt confident that my changes to the schedule were saved correctly.	☐	☐	☐	☐	☐

What parts of the scheduling process felt unclear or unintuitive (if any)?

Are there any real classroom or school constraints that might make scheduling quizzes difficult using this system?

Scenario 4

Scenario 4: Quiz Attempt (Student View)

You will now switch to the perspective of a student using the mobile platform to attempt quizzes.

Your task

View the three videos below that show attempts of the three quiz types: a Basic Quiz, a Crossword Quiz, and a Rapid Quiz.

After viewing all three videos, move on to the next section and answer the questions that follow.
You do not need to do anything besides watching the videos.
If you are unable to open the videos in full-screen, please view them directly on youtube by clicking the "Youtube" button on the navigation bar of each video.

(Videos are embedded in actual survey)

Basic Quiz Attempt
https://www.youtube.com/watch?v=T1xeg_nD7F8

Crossword Quiz Attempt
https://www.youtube.com/watch?v=lemNPc8x4Kw

Rapid Quiz Attempt
https://www.youtube.com/watch?v=fcI0ug-0_ww

Scenario 4 Review

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The mobile application could be used independently by students after minimal initial exposure.	☐	☐	☐	☐	☐
The application design is well-suited for short revision sessions.	☐	☐	☐	☐	☐
The quiz experience is engaging for students.	☐	☐	☐	☐	☐
Time pressure during quizzes (where applicable) was perceived as appropriate rather than stressful.	☐	☐	☐	☐	☐
A mobile application increases student accessibility to the platform, compared to a web-based platform.	☐	☐	☐	☐	☐
Gamification features (leaderboards, scores, rewards, avatar customization) would increase student engagement.	☐	☐	☐	☐	☐
The mobile application feels well-suited for brief, on-the-go use.	☐	☐	☐	☐	☐

What parts of the student quiz experience might confuse a primary school student?

What factors might prevent students from using this application regularly in practice?

Perceived Value

In the following questions, the platform refers to both the website and mobile application as a whole. Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The platform feels practical for day-to-day classroom use.	☐	☐	☐	☐	☐
The platform integrates well into my regular teaching routine.	☐	☐	☐	☐	☐
Using this platform would save me time compared to my current revision practices.	☐	☐	☐	☐	☐
Managing revision through this platform would feel more efficient.	☐	☐	☐	☐	☐
The platform reduces the administrative burden of organising revision activities.	☐	☐	☐	☐	☐
The platform provides clear value beyond my current tools.	☐	☐	☐	☐	☐
The platform fits within typical school constraints (time, resources).	☐	☐	☐	☐	☐
The platform encourages students to engage in independent, self-directed learning.	☐	☐	☐	☐	☐

AI Quiz Generation

The system is intended to support AI-assisted quiz generation in the future. Please answer the following questions based on your experience creating quizzes during the session.

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
Use of AI-generated quizzes would reduce the effort required to create revision questions.	☐	☐	☐	☐	☐
AI-generated questions would be considered trustworthy after review.	☐	☐	☐	☐	☐
Use of AI-generated quizzes for revision would feel comfortable and acceptable.	☐	☐	☐	☐	☐
Control over topic coverage should be available when using AI-generated quizzes.	☐	☐	☐	☐	☐
AI-generated quizzes should align closely with the syllabus.	☐	☐	☐	☐	☐

What aspects would you want AI to assist with

☐ Generating questions

☐ Generating answer options

☐ Balancing difficulty

☐ Covering syllabus topics

☐ Formatting / structuring quizzes

Concluding Remarks

Any other feedback